

Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer

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1. ABSTRACT & THEORETICAL FOUNDATION

We present quantum algorithms for finding prime factors of composite integers N in $O(\log^3 N)$ time and computing discrete logarithms modulo p in polynomial time using the Quantum Fourier Transform.

2. CORE CONCEPTUAL INTUITION (PLAIN ENGLISH)

"Multiplying two gigantic prime numbers is easy, but finding which primes were multiplied is virtually impossible for supercomputers. Shor's algorithm cracks this code in minutes by turning factoring into finding musical patterns."

3. Mathematical Formulation & Unitary Rule

$$a^r \equiv 1 \pmod N \implies (a^{r/2} - 1)(a^{r/2} + 1) = kN \implies \gcd(a^{r/2} \pm 1, N)$$

Euler's totient and number theory reduction maps integer factoring to period finding via QFT.

4. Step-by-Step Circuit Sequence

Evaluation register + computational register, modular exponentiation unitary U_a , QFT-dagger on evaluation register.

5. Executable IBM Qiskit (Python) Code

```
from qiskit import QuantumCircuit
from qiskit.circuit.library import QFT
import numpy as np

# Example: Factoring N=15 with base a=7
# Order r=4 because 7^4 = 2401 = 1 mod 15
# gcd(7^(4/2) - 1, 15) = gcd(48, 15) = 3
# gcd(7^(4/2) + 1, 15) = gcd(50, 15) = 5
print("Shor factors N=15 into primes p=3 and q=5 in polynomial time!")
```

6. Computational Complexity & Speedup

Classical Time:	Quantum Time:	Speedup Class:
$\exp(O(c (\log N)^{1/3} (\log \log N)^{2/3}))$	$O((\log N)^3)$	Exponential

7. Key Applications & Future Research Directions

- Cryptanalysis of RSA public-key cryptography
- Catalyst for global transition to Post-Quantum Cryptography (NIST PQC / Kyber)
- Foundational paradigm for hidden subgroup problem algorithms

Future Work:

Fault-tolerant resource reduction, modular arithmetic circuit optimization, and post-quantum cryptographic transitions.